

Lab 1 Use Case Flow

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Problem 26 Peer to Peer Carpooling and Expense Splitter

Goal	Confirm Booking
Actor	Rider Commuter with help from Driver Host and Payment Gateway
Main flow	1. Rider picks a route from the list and taps Book Seat. 2. System shows route details: pickup, drop, time and detour. 3. System calculates fuel cost share and shows it. This includes UC-05 Calculate Fuel Cost Split. 4. Rider checks the cost and confirms the booking. 5. System sends the amount to Payment Gateway. This includes UC-06 Make Payment. 6. Payment Gateway approves the payment. 7. System reduces free seats by one. 8. System sends booking confirmation to rider and driver. 9. Use case ends.
Alternate Flow	Payment Declined At step 6, if payment fails: 6a1. Payment Gateway sends decline. 6a2. System shows Payment failed and does not reduce seats. 6a3. Rider can try again with same or other payment method. 6a4. If it fails again after 2 tries, booking is cancelled and seat is freed. Rider is notified. Seat Not Available At step 7, if seat is taken: 7a1. Another rider books the last seat first. 7a2. System finds conflict and shows Seat not available. 7a3. Rider goes back to results to pick another route. No payment is taken.

Preconditions	1. Rider is logged in. 2. A matching route is shown within 1.5 km detour limit. 3. Driver has at least one free seat.
Postconditions	Success: Seat is booked, cost share is shown, payment is taken, and both rider and driver get confirmation. Failure: No seat is booked and no payment is taken. Rider goes back to results.